

Yuni Wu

South San Francisco, CA

yunihsuanwu@gmail.com | (720) 487-5403 | linkedin/wih | github/yuni-wyx | portfolio

Summary

Machine Learning Engineer focused on dependable ML systems, not just promising model scores. Builds evaluation and deployment-validation workflows that test whether model improvements hold up under class imbalance, latency, and inference constraints.

Experience

Graduate Research Assistant

Sept 2025 – May 2026

CAIRO Lab, University of Colorado Boulder — Embodied AI and robotics research lab

- Curated 97K+ Unitree Go1 state-action samples with ROS/rosbag pipelines for reproducible offline evaluation.
- Constructed and benchmarked an offline evaluation dataset containing 2.4K+ labeled state-action sequences across 13 robot-navigation scenarios to assess VLM decision-making consistency.

Machine Learning Engineer Intern

May 2025 – Aug 2025

MyEdMaster LLC — AI healthcare and education startup

- Achieved 0.92 F1-score with a PyTorch/torchvision ResNet classifier on 50K+ medical image samples.
- Improved minority-class F1 by +30% with hierarchical classification, targeted oversampling, and imbalance analysis.
- Engineered evaluation pipelines with cross-validation, ROC-AUC, calibration, and confusion-matrix analysis to assess model reliability and identify prediction failure modes under class imbalance.
- Reduced inference latency by $5\times$ (0.15s to 0.03s per image) through ResNet ablation and deployment tradeoff analysis.

Software Engineer

June 2021 – Aug 2022

Shopee Co. Ltd. — Leading Southeast Asian e-commerce platform

- Reduced QA workload by 80% by engineering validation and anomaly-detection pipelines on 800+ QA scenarios.
- Accelerated reporting speed by 78% with regression monitoring pipelines that surfaced data inconsistencies earlier.
- Reduced query latency by 15% by optimizing distributed ETL workflows and GCP data-access patterns.

Projects

NailAI — Cloud-Native Inference Platform

[demo]

- Evaluated cloud-native image classification with PyTorch/torchvision ResNet inference workflows.
- Enabled asynchronous image processing through Cloud Run, Pub/Sub, Cloud Storage, idempotent orchestration, and retries.
- Achieved 0% failure at 200 concurrent users with p95 latency of 6.8s while preserving request/prediction metadata.

VLM-Based Social Navigation for Quadruped

[project]

- Trained and deployed PyTorch/torchvision ResNet-based behavior-cloning policies on 97K+ state-action samples.
- Achieved 0.11 MAE and $R^2 = 0.62$ in waypoint prediction under trajectory-split offline evaluation.
- Compared InternVL, Qwen-VL, and prompt strategies across 2.4K+ labeled sequences and 13 scenarios.

Technical Skills

Languages: Python, SQL, Go, R

Machine Learning: PyTorch, torchvision, NumPy, scikit-learn evaluation metrics, CNNs, ResNet, Transformers

Evaluation: cross-validation, ROC-AUC, calibration, F1-score, class imbalance, benchmarking, ablation workflows

Deployment: Cloud Run inference evaluation, Docker, Cloud Storage, asynchronous processing, monitoring, structured logging

Data & Systems: BigQuery, PostgreSQL, ETL/ELT, validation workflows, anomaly detection, regression monitoring

Education

University of Colorado Boulder — MS, Data Science | GPA: 3.76/4.0

May 2026

Soochow University — BS, Data Science | GPA: 3.7/4.0

June 2021